```
> # DISCOVERY OF IDENTITIES FOR a_4(n)
  with(qseries) :
  T := 60 : N := 7 * T + 20 :
  f := d → etaq(q, d, N) :
  tq := q * f(7)^4 / f(1)^4 :
  p0q := q * f(14)^4 * f(1)^4 / (f(7)^4 * f(2)^4) :
  p1q := (f(14) * f(1)^7 / (f(7) * f(2)^7) − 1) / 7 :
  A4q := f(2)^3 * f(49)^4 / (q^4 * f(1)^4 * f(98)^3) :
  B4q := 1 :

  # GROUP 1, IDENTITY 1: U_A4(p0)
  m := 5 :
  S4_1_1 := series(A4q * p0q, q, 7 * T) :
  target4_1_1 := add(coeff(S4_1_1, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_1_1 := expand(findlincombo(target4_1_1, basis4, symbols4, q, 0)) :
  collect(R4_1_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$(7)^2\, t^2\, p0 + (7)\, t \qquad \textbf{(1)}$$

```
> # GROUP 1, IDENTITY 2: U_A4(p0*t^(-1))
  m := 5 :
  S4_1_2 := series(A4q * p0q / tq, q, 7 * T) :
  target4_1_2 := add(coeff(S4_1_2, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_1_2 := expand(findlincombo(target4_1_2, basis4, symbols4, q, 0)) :
  collect(R4_1_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$-(7)\, t\, p0 + p1 - 1 - (7)\, t \qquad \textbf{(2)}$$

```
> # GROUP 1, IDENTITY 3: U_A4(p0*t^(-2))
  m := 5 :
  S4_1_3 := series(A4q * p0q / tq^2, q, 7 * T) :
  target4_1_3 := add(coeff(S4_1_3, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
```

```
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_1_3 := expand(findlincombo(target4_1_3, basis4, symbols4, q, 0)) :
collect(R4_1_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left((7)^5 t^3 + (2)\,(7)^4 t^2 + (2)\,(3)^3\,(7)\,t\right) p0 + \left(-(7)^4 t^2 - (7)^2\,(13)\,t - (2)^4\,(3)\right) p1 + (7)^4 t^2 + (5)\,(7)\,(19)\,t + (2)^4\,(3) \tag{3}$$

```
> # GROUP 1, IDENTITY 4: U_A4(p0*t^(-3))
m := 5 :
S4_1_4 := series(A4q * p0q / tq^3, q, 7 * T) :
target4_1_4 := add(coeff(S4_1_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_1_4 := expand(findlincombo(target4_1_4, basis4, symbols4, q, 0)) :
collect(R4_1_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left((7)^7 t^4 + (2)\,(7)^5 t^3 - (7)^4\,(13)\,t^2 - (3)^3\,(7)\,(17)\,t\right) p0 + \left(-(7)^6 t^3 - (7)^4 t^2 + (3)\,(7)^2\,(29)\,t + (2)\,(7)\,(29)\right) p1 + (5)\,(7)^5 t^3 - (3)\,(7)^3 t^2 - (2)\,(3)^2\,(5)\,(7)^2 t - (2)\,(7)\,(29) \tag{4}$$

```
> # GROUP 1, IDENTITY 5: U_A4(p0*t^(-4))
m := 6 :
S4_1_5 := series(A4q * p0q / tq^4, q, 7 * T) :
target4_1_5 := series(tq * (add(coeff(S4_1_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
    T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_1_5 := expand(findlincombo(target4_1_5, basis4, symbols4, q, 0) / t) :
collect(R4_1_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms , 42$$

$$\left(-(7)^9 t^5 - (2)^2\,(3)^2\,(7)^7 t^4 - (2)\,(7)^6\,(19)\,t^3 - (7)^3\,(181)\,t^2 + (3)\,(7)^2\,(11)^2\,t + (2)^3\right) p0 + \left((7)^8 t^4 + (5)\,(7)^7 t^3 + (2)\,(7)^5\,(17)\,t^2 + (2)^4\,(7)^3\,t - (3)\,(7)\,(107) - \frac{1}{t}\right) p1 - (7)^8 t^4 - (7)^6\,(31)\,t^3 - (2)\,(7)^4\,(109)\,t^2 - (5)\,(7)^2\,(19)\,t + (3)\,(7)\,(107) \tag{5}$$

$$+\frac{1}{t}$$

```
> # GROUP 1, IDENTITY 6: U_A4(p0*t^(-5))
  m := 7 :
  S4_1_6 := series(A4q * p0q / tq^5, q, 7 * T) :
  target4_1_6 := series(tq * (add(coeff(S4_1_6, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_1_6 := expand(findlincombo(target4_1_6, basis4, symbols4, q, 0) / t) :
  collect(R4_1_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\begin{aligned}&\left((7)^{11}\,t^6+(7)^9\,(41)\,t^5+(2)\,(7)^7\,(367)\,t^4+(7)^6\,(727)\,t^3+(2)\,(3)^2\,(7)^4\,(89)\,t^2\right.\\&\left.-(5)^2\,(7)^2\,(41)\,t-(2)^5\,(5)\right)p0+\left(-(7)^{10}\,t^5-(2)^3\,(5)\,(7)^8\,t^4-(2)^2\,(5)^2\,(7)^7\,t^3\right.\\&\left.-(7)^5\,(659)\,t^2-(2)\,(3)\,(7)^3\,(229)\,t+(7)\,(11)\,(83)+\frac{(2)^2\,(5)}{t}\right)p1\\&+(2)\,(3)\,(7)^9\,t^5+(5)\,(7)^7\,(53)\,t^4+(3)\,(7)^7\,(31)\,t^3+(2)^2\,(3)\,(7)^4\,(367)\,t^2\\&+(3)\,(7)^2\,(19)\,(167)\,t-(2)\,(3)^2\,(5)\,(71)-\frac{(2)^2\,(5)}{t}\end{aligned}\tag{6}$$

```
> # GROUP 1, IDENTITY 7: U_A4(p0*t^(-6))
  m := 8 :
  S4_1_7 := series(A4q * p0q / tq^6, q, 7 * T) :
  target4_1_7 := series(tq * (add(coeff(S4_1_7, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_1_7 := expand(findlincombo(target4_1_7, basis4, symbols4, q, 0) / t) :
  collect(R4_1_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$

$$\#\ of\ terms\ , 48$$

$$\begin{aligned}&\left(-(7)^{13}\,t^7-(2)\,(7)^{11}\,(23)\,t^6-(7)^9\,(947)\,t^5-(2)^{10}\,(7)^7\,(11)\,t^4-(7)^6\,(11)\,(877)\,t^3\right.\\&\left.-(7)^4\,(59)\,(401)\,t^2-(2)\,(7)^2\,(31)\,(107)\,t+(2)^3\,(11)\,(23)\right)p0+\left((7)^{12}\,t^6\right.\\&\left.+(3)^2\,(5)\,(7)^{10}\,t^5+(7)^8\,(907)\,t^4+(2)^3\,(3)^3\,(7)^8\,t^3+(7)^5\,(8747)\,t^2\right.\end{aligned}\tag{7}$$

$$+ (2)\,(7)^3\,(10343)\,t + (2)^4\,(7)\,(359) - \frac{(11)\,(23)}{t}\Big)\,p1 - (5)\,(7)^{11}\,t^6$$
$$- (2)\,(3)\,(7)^9\,(43)\,t^5 - (2)^2\,(3)^2\,(7)^8\,(23)\,t^4 - (7)^7\,(1423)\,t^3 - (2)^2\,(7)^4\,(14813)\,t^2$$
$$- (2)\,(7)^2\,(72109)\,t - (7)^2\,(821) + \frac{(11)\,(23)}{t}$$

```
> # GROUP 2, IDENTITY 1: U_A4(p1)
  m := 8 :
  S4_2_1 := series(A4q * p1q, q, 7 * T) :
  target4_2_1 := add(coeff(S4_2_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_2_1 := expand(findlincombo(target4_2_1, basis4, symbols4, q, 0)) :
  collect(R4_2_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$
$$\#\ of\ terms\ , 48$$

$$\big((2)^3\,(7)^{11}\,t^7 + (2)^4\,(3)^2\,(7)^9\,t^6 + (7)^8\,(139)\,t^5 + (2)^2\,(7)^6\,(101)\,t^4 + (3)\,(7)^4\,(139)\,t^3 - (3)\,(5)\,(7)^2\,t^2 - (2)^3\,(7)\,t\big)\,p0 + \big((7)^{14}\,t^8 + (3)^3\,(7)^{12}\,t^7 + (7)^{10}\,(293)\,t^6 + (2)^2\,(7)^8\,(401)\,t^5 + (2)\,(7)^7\,(17)\,(19)\,t^4 + (5)\,(7)^5\,(13)^2\,t^3 + (2)^3\,(7)^5\,t^2 + (2)\,(3)\,(7)\,(19)\,t + (7)\big)\,p1 + (7)^{13}\,t^8 + (7)^{11}\,(19)\,t^7 + (3)\,(7)^9\,(47)\,t^6 + (3)\,(7)^8\,(23)\,t^5 + (3)^2\,(7)^6\,(11)\,t^4 - (2)^2\,(3)\,(7)^4\,t^3 - (3)^3\,(7)^2\,(11)\,t^2 - (2)^3\,(7)\,(13)\,t - (7) \qquad (8)$$

```
> # GROUP 2, IDENTITY 2: U_A4(p1*t^(-1))
  m := 5 :
  S4_2_2 := series(A4q * p1q / tq, q, 7 * T) :
  target4_2_2 := add(coeff(S4_2_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_2_2 := expand(findlincombo(target4_2_2, basis4, symbols4, q, 0)) :
  collect(R4_2_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$-(2)^3\,(7)\,t\,p0 + \big((7)^2\,t + (7)\big)\,p1 - (7) \qquad (9)$$

```
> # GROUP 2, IDENTITY 3: U_A4(p1*t^(-2))
  m := 5 :
  S4_2_3 := series(A4q * p1q / tq^2, q, 7 * T) :
  target4_2_3 := add(coeff(S4_2_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
```

```
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_2_3 := expand(findlincombo(target4_2_3, basis4, symbols4, q, 0)) :
collect(R4_2_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$((3)\ (7)^3\ t^2 + (7)\ (11)\ t)\ p0 + (-(3)\ (7)^2\ t - (3)^2)\ p1 + (7)^3\ t^2 + (2)\ (3)^2\ (7)\ t + (3)^2 \quad (10)$$

```
> # GROUP 2, IDENTITY 4: U_A4(p1*t^(-3))
  m := 5 :
  S4_2_4 := series(A4q * p1q / tq^3, q, 7 * T) :
  target4_2_4 := add(coeff(S4_2_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_2_4 := expand(findlincombo(target4_2_4, basis4, symbols4, q, 0)) :
  collect(R4_2_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$((2)\ (7)^5\ t^3 + (3)^3\ (7)^3\ t^2 + (7)\ (131)\ t)\ p0 + (-(2)\ (7)^4\ t^2 - (2)^3\ (3)\ (7)^2\ t - (2)\ (59))\ p1 - (7)^5\ t^3 + (2)^2\ (7)^3\ t^2 + (5)^2\ (7)^2\ t + (2)\ (59) \quad (11)$$

```
> # GROUP 2, IDENTITY 5: U_A4(p1*t^(-4))
  m := 5 :
  S4_2_5 := series(A4q * p1q / tq^4, q, 7 * T) :
  target4_2_5 := series(tq * (add(coeff(S4_2_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_2_5 := expand(findlincombo(target4_2_5, basis4, symbols4, q, 0) / t) :
  collect(R4_2_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$((2)\ (7)^7\ t^4 - (2)^8\ (7)^3\ t^2 - (2)\ (7)^2\ (103)\ t - (2)^3)\ p0 + \left(-(2)\ (7)^6\ t^3 + (3)\ (7)^4\ t^2 + (7)^2\ (241)\ t + (3)\ (7)\ (61) + \frac{1}{t}\right) p1 - (7)^7\ t^4 + (2)^2\ (7)^5\ t^3 - (7)^3\ (23)\ t^2 - (7)^2\ (13)\ (19)\ t - (3)\ (7)\ (61) - \frac{1}{t} \quad (12)$$

```
> # GROUP 2, IDENTITY 6: U_A4(p1*t^(-5))
  m := 6 :
  S4_2_6 := series(A4q * p1q / tq^5, q, 7 * T) :
  target4_2_6 := series(tq * (add(coeff(S4_2_6, q, 7 * n) * q^n, n = -1 .. T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q
      * tq^j, q, T), j = 0 .. m)] :
  symbols4 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
  R4_2_6 := expand(findlincombo(target4_2_6, basis4, symbols4, q, 0) / t) :
  collect(R4_2_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&\left(-(5)\,(7)^9\,t^5-(2)\,(3)\,(7)^7\,(23)\,t^4-(7)^6\,(11)\,(13)\,t^3-(3)^2\,(7)^4\,(19)\,t^2\right.\\&\quad\left.+(2)^3\,(7)^2\,(149)\,t+(2)^3\,(19)\right)p0+\left((5)\,(7)^8\,t^4+(7)^7\,(19)\,t^3+(7)^5\,(127)\,t^2\right.\\&\quad\left.+(2)\,(5)\,(7)^3\,(13)\,t-(2)\,(7)\,(23)^2-\frac{(19)}{t}\right)p1-(2)\,(7)^7\,(11)\,t^4\\&\quad-(2)\,(7)^6\,(53)\,t^3-(2)\,(7)^4\,(11)\,(37)\,t^2-(2)\,(7)^2\,(431)\,t+(3)^2\,(823)+\frac{(19)}{t}\end{aligned}\tag{13}$$

```
> # GROUP 2, IDENTITY 7: U_A4(p1*t^(-6))
  m := 7 :
  S4_2_7 := series(A4q * p1q / tq^6, q, 7 * T) :
  target4_2_7 := series(tq * (add(coeff(S4_2_7, q, 7 * n) * q^n, n = -1 .. T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q
      * tq^j, q, T), j = 0 .. m)] :
  symbols4 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
  R4_2_7 := expand(findlincombo(target4_2_7, basis4, symbols4, q, 0) / t) :
  collect(R4_2_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\begin{aligned}&\left((2)^3\,(7)^{11}\,t^6+(2)\,(7)^{10}\,(19)\,t^5+(2)\,(3)^4\,(7)^7\,(23)\,t^4+(2)\,(7)^6\,(17)\,(97)\,t^3\right.\\&\quad\left.+(7)^4\,(13)\,(571)\,t^2-(7)^2\,(2693)\,t-(2)^3\,(233)\right)p0+\left(-(2)^3\,(7)^{10}\,t^5\right.\\&\quad-(7)^9\,(37)\,t^4-(2)^3\,(3)^2\,(7)^8\,t^3-(2)\,(7)^5\,(1489)\,t^2-(2)^2\,(7)^3\,(1601)\,t\\&\quad\left.+(7)\,(11)\,(13)\,(17)+\frac{(233)}{t}\right)p1+(7)^{11}\,t^6+(2)^2\,(7)^9\,(17)\,t^5+(7)^7\,(11)\,(157)\,t^4\\&\quad+(2)\,(7)^6\,(1619)\,t^3+(2)^2\,(7)^4\,(4969)\,t^2+(7)^4\,(907)\,t-(17029)-\frac{(233)}{t}\end{aligned}\tag{14}$$

```
> # GROUP 3, IDENTITY 1: U_A4(1)
  m := 5 :
  S4_3_1 := series(A4q, q, 7 * T) :
  target4_3_1 := add(coeff(S4_3_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_3_1 := expand(findlincombo(target4_3_1, basis4, symbols4, q, 0)) :
  collect(R4_3_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\begin{aligned}&\left((3)\,(7)^8\,t^5+(2)^2\,(7)^6\,(13)\,t^4+(3)\,(7)^4\,(107)\,t^3+(3)\,(7)^3\,(37)\,t^2+(2)^3\,(3)^2\,(7)\,t\right)p0\\&+\left(-(3)\,(7)^7\,t^4-(2)\,(5)^2\,(7)^5\,t^3-(3)^3\,(7)^3\,(11)\,t^2-(3)^2\,(7)^2\,(11)\,t\right.\\&\left.-(3)^2\,(7)\right)p1+(7)^8\,t^5+(2)\,(7)^6\,(17)\,t^4+(3)^4\,(5)\,(7)^4\,t^3+(2)\,(5)^2\,(7)^2\,(43)\,t^2\\&+(5)\,(7)\,(139)\,t+(3)^2\,(7)\end{aligned}\qquad\textbf{(15)}$$

```
> # GROUP 3, IDENTITY 2: U_A4(t^(-1))
  m := 5 :
  S4_3_2 := series(A4q / tq, q, 7 * T) :
  target4_3_2 := add(coeff(S4_3_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_3_2 := expand(findlincombo(target4_3_2, basis4, symbols4, q, 0)) :
  collect(R4_3_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$-(7)\,t\qquad\textbf{(16)}$$

```
> # GROUP 3, IDENTITY 3: U_A4(t^(-2))
  m := 5 :
  S4_3_3 := series(A4q / tq^2, q, 7 * T) :
  target4_3_3 := add(coeff(S4_3_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_3_3 := expand(findlincombo(target4_3_3, basis4, symbols4, q, 0)) :
  collect(R4_3_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((3)\,(7)^3\,t^2+(7)\,(11)\,t\right)p0+\left(-(3)\,(7)^2\,t-(2)\,(5)\right)p1+(7)^3\,t^2+(2)\,(7)\,(13)\,t\qquad\textbf{(17)}$$

$+ (2)\ (5)$

```
> # GROUP 3, IDENTITY 4: U_A4(t^(-3))
  m := 5 :
  S4_3_4 := series(A4q / tq^3, q, 7 * T) :
  target4_3_4 := series(tq * (add(coeff(S4_3_4, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_3_4 := expand(findlincombo(target4_3_4, basis4, symbols4, q, 0) / t) :
  collect(R4_3_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left(-(2)\ (3)\ (7)^5\ t^3 - (5)\ (7)^3\ (17)\ t^2 - (5)\ (7)^3\ t + (2)^3\right) p0 + \left((2)\ (3)\ (7)^4\ t^2 + (2)^4\ (5)\ (7)^2\ t + (7)\ (31) - \frac{1}{t}\right) p1 - (7)^5\ t^3 - (2)^2\ (7)^3\ (13)\ t^2 - (3)^2\ (5)\ (7)\ (13)\ t - (7)\ (31) + \frac{1}{t} \tag{18}$$

```
> # GROUP 3, IDENTITY 5: U_A4(t^(-4))
  m := 5 :
  S4_3_5 := series(A4q / tq^4, q, 7 * T) :
  target4_3_5 := series(tq * (add(coeff(S4_3_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_3_5 := expand(findlincombo(target4_3_5, basis4, symbols4, q, 0) / t) :
  collect(R4_3_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left(-(2)\ (3)\ (7)^7\ t^4 - (2)^3\ (7)^5\ t^3 + (2)^4\ (5)\ (7)^4\ t^2 + (2)\ (3)^2\ (7)^2\ (17)\ t - (2)^5\ (3)\right) p0 + \left((2)\ (3)\ (7)^6\ t^3 + (2)^2\ (7)^4\ t^2 - (2)^2\ (7)^3\ (19)\ t - (2)\ (3)^3\ (5)\ (7) + \frac{(2)^2\ (3)}{t}\right) p1 - (7)^7\ t^4 - (2)^2\ (3)^2\ (7)^5\ t^3 + (2)\ (5)^2\ (7)^3\ t^2 + (2)\ (5)^2\ (7)^2\ (11)\ t + (2)\ (3)^3\ (5)\ (7) - \frac{(2)^2\ (3)}{t} \tag{19}$$

```
> # GROUP 3, IDENTITY 6: U_A4(t^(-5))
```

```
m := 6 :
S4_3_6 := series(A4q/tq^5, q, 7*T) :
target4_3_6 := series(tq*(add(coeff(S4_3_6, q, 7*n)*q^n, n=-1..T-1) + O(q^T)), q,
    T) :
basis4 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)] :
symbols4 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
R4_3_6 := expand(findlincombo(target4_3_6, basis4, symbols4, q, 0)/t) :
collect(R4_3_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&((3)\,(7)^9\,t^5 + (2)\,(7)^7\,(79)\,t^4 + (7)^7\,(23)\,t^3 - (7)^4\,(67)\,t^2 - (2)^4\,(3)^2\,(7)^2\,(13)\,t \\ &\quad + (2)^3\,(101))\,p0 + \left(-(3)\,(7)^8\,t^4 - (2)\,(7)^7\,(11)\,t^3 - (5)\,(7)^5\,(29)\,t^2\right. \\ &\quad \left. + (2)^2\,(7)^3\,(23)\,t + (2)\,(3)\,(5)^2\,(7)\,(11) - \frac{(101)}{t}\right)p1 + (2)\,(7)^7\,(17)\,t^4 \\ &\quad + (7)^6\,(149)\,t^3 + (2)\,(3)\,(5)\,(7)^4\,(31)\,t^2 - (2)^2\,(3)\,(7)^2\,(61)\,t - (11549) + \frac{(101)}{t}\end{aligned} \tag{20}$$

```
> # GROUP 3, IDENTITY 7: U_A4(t^(-6))
m := 7 :
S4_3_7 := series(A4q/tq^6, q, 7*T) :
target4_3_7 := series(tq*(add(coeff(S4_3_7, q, 7*n)*q^n, n=-1..T-1) + O(q^T)), q,
    T) :
basis4 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)] :
symbols4 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
R4_3_7 := expand(findlincombo(target4_3_7, basis4, symbols4, q, 0)/t) :
collect(R4_3_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\begin{aligned}&(-(2)\,(5)\,(7)^{10}\,t^5 - (2)\,(5)^2\,(7)^9\,t^4 - (2)\,(7)^6\,(17)\,(83)\,t^3 - (5)\,(7)^4\,(1109)\,t^2 \\ &\quad + (3)\,(7)^2\,(2393)\,t - (2)^5\,(5)^2\,(7))\,p0 + \left((2)^2\,(7)^8\,(17)\,t^4 + (2)\,(7)^7\,(167)\,t^3\right. \\ &\quad \left. + (2)^9\,(5)\,(7)^5\,t^2 + (7)^3\,(4721)\,t - (2)^6\,(3)^2\,(7)\,(11) + \frac{(2)^2\,(5)^2\,(7)}{t}\right)p1 \\ &\quad + (7)^{11}\,t^6 + (2)^2\,(5)\,(7)^9\,t^5 - (2)\,(7)^7\,(13)\,(17)\,t^4 - (2)^2\,(7)^6\,(13)\,(43)\,t^3 \\ &\quad - (2)\,(7)^4\,(8609)\,t^2 - (2)^7\,(3)\,(5)\,(7)^2\,(17)\,t + (2)^2\,(11083) - \frac{(2)^2\,(5)^2\,(7)}{t}\end{aligned} \tag{21}$$

```
> # GROUP 4, IDENTITY 1: U_B4(p0)
```

```
m := 5 :
S4_4_1 := series(B4q * p0q, q, 7 * T) :
target4_4_1 := add(coeff(S4_4_1, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_4_1 := expand(findlincombo(target4_4_1, basis4, symbols4, q, 0)) :
collect(R4_4_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^3\ (7)^7\ t^4 + (2)^4\ (5)\ (7)^5\ t^3 + (2)^3\ (3)^3\ (7)^3\ t^2 + (7)\ (79)\ t\right) p0 + \left((7)^{10}\ t^5 + (7)^8\ (19)\ t^4 + (2)\ (3)^2\ (7)^7\ t^3 + (3)\ (7)^4\ (109)\ t^2 + (2)\ (7)^3\ (17)\ t + (2)^2\right) p1 + (7)^9\ t^5 + (7)^7\ (11)\ t^4 + (2)\ (7)^5\ (19)\ t^3 + (7)^3\ (31)\ t^2 + (2)\ (3)\ (7)\ t \tag{22}$$

```
> # GROUP 4, IDENTITY 2: U_B4(p0*t^(-1))
m := 5 :
S4_4_2 := series(B4q * p0q / tq, q, 7 * T) :
target4_4_2 := add(coeff(S4_4_2, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_4_2 := expand(findlincombo(target4_4_2, basis4, symbols4, q, 0)) :
collect(R4_4_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^3\ t^2 - (2)^3\ (7)\ t\right) p0 + \left((7)^2\ t + (7)\right) p1 + 1 - (7)^2\ t \tag{23}$$

```
> # GROUP 4, IDENTITY 3: U_B4(p0*t^(-2))
m := 5 :
S4_4_3 := series(B4q * p0q / tq^2, q, 7 * T) :
target4_4_3 := add(coeff(S4_4_3, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_4_3 := expand(findlincombo(target4_4_3, basis4, symbols4, q, 0)) :
collect(R4_4_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^5\ t^3 - (2)^3\ (7)^3\ t^2\right) p0 + \left((7)^4\ t^2 + (7)^3\ t\right) p1 - (7)^4\ t^2 - (7)^3\ t - (2)^2\ (3) \tag{24}$$

```
> # GROUP 4, IDENTITY 4: U_B4(p0*t^(-3))
m := 5 :
```

```
S4_4_4 := series(B4q * p0q / tq^3, q, 7 * T) :
target4_4_4 := add(coeff(S4_4_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_4_4 := expand(findlincombo(target4_4_4, basis4, symbols4, q, 0)) :
collect(R4_4_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(1 + (2)\,(3)\,(7)^7\,t^4 + (2)\,(7)^5\,(31)\,t^3 + (3)^3\,(7)^4\,t^2 + (2)\,(5)\,(7)^2\,t\right) p0 + \left(\tag{25}$$
$$-(2)\,(3)\,(7)^6\,t^3 - (2)^3\,(7)^5\,t^2 - (3)\,(7)^4\,t - (2)^2\,(7)\right) p1 + (2)\,(7)^6\,t^3$$
$$+ (2)^3\,(3)\,(7)^4\,t^2 + (7)^2\,(83)\,t + (2)^2\,(3)^3$$

```
> # GROUP 4, IDENTITY 5: U_B4(p0*t^(-4))
m := 5 :
S4_4_5 := series(B4q * p0q / tq^4, q, 7 * T) :
target4_4_5 := add(coeff(S4_4_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_4_5 := expand(findlincombo(target4_4_5, basis4, symbols4, q, 0)) :
collect(R4_4_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^9\,t^5 - (2)\,(5)^2\,(7)^7\,t^4 - (2)^2\,(3)\,(5)\,(7)^6\,t^3 - (2)\,(3)^4\,(7)^4\,t^2 - (2)^2\,(3)\,(5)\,(7)^2\,t \right. \tag{26}$$
$$\left. - (2)\,(3)\right) p0 + \left((7)^8\,t^4 + (7)^8\,t^3 + (2)\,(3)^3\,(7)^5\,t^2 + (2)\,(3)^2\,(7)^4\,t\right.$$
$$\left. + (2)^3\,(3)\,(7)\right) p1 - (7)^8\,t^4 - (5)^2\,(7)^6\,t^3 - (2)\,(3)\,(7)^4\,(31)\,t^2 - (2)\,(3)\,(7)^2\,(83)\,t$$
$$- (2)^3\,(3)\,(5)\,(7)$$

```
> # GROUP 4, IDENTITY 6: U_B4(p0*t^(-5))
m := 6 :
S4_4_6 := series(B4q * p0q / tq^5, q, 7 * T) :
target4_4_6 := add(coeff(S4_4_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_4_6 := expand(findlincombo(target4_4_6, basis4, symbols4, q, 0)) :
collect(R4_4_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left(-(7)^{11}\,t^6 - (2)^3\,(7)^9\,t^5 + (3)\,(7)^9\,t^4 + (2)\,(3)\,(5)\,(7)^7\,t^3 + (3)^4\,(7)^5\,t^2 \right. \tag{27}$$

$$+ (2)\,(3)\,(5)\,(7)^3\,t + (3)\,(7))\,p0 + ((7)^{10}\,t^5 + (7)^9\,t^4 - (3)\,(7)^8\,t^3 - (3)^3\,(7)^6\,t^2$$
$$- (3)^2\,(7)^5\,t - (2)^2\,(3)\,(7)^2)\,p1 - (7)^{10}\,t^5 - (7)^9\,t^4 + (3)^2\,(7)^7\,t^3 + (3)\,(7)^5\,(31)\,t^2$$
$$+ (3)\,(7)^3\,(83)\,t + (2)^2\,(7)\,(11)\,(19)$$

```
> # GROUP 4, IDENTITY 7: U_B4(p0*t^(-6))
  m := 7 :
  S4_4_7 := series(B4q * p0q / tq^6, q, 7 * T) :
  target4_4_7 := add(coeff(S4_4_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_4_7 := expand(findlincombo(target4_4_7, basis4, symbols4, q, 0)) :
  collect(R4_4_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$
$$\#\ of\ terms\ , 45$$

$$(-(7)^{13}\,t^7 - (2)^3\,(7)^{11}\,t^6 + (2)\,(7)^9\,t^4 + (2)^2\,(5)\,(7)^7\,t^3 + (2)\,(3)^3\,(7)^5\,t^2 \qquad (28)$$
$$+ (2)^2\,(5)\,(7)^3\,t + (2)\,(7))\,p0 + ((7)^{12}\,t^6 + (7)^{11}\,t^5 - (2)\,(7)^8\,t^3 - (2)\,(3)^2\,(7)^6\,t^2$$
$$- (2)\,(3)\,(7)^5\,t - (2)^3\,(7)^2)\,p1 - (7)^{12}\,t^6 - (7)^{11}\,t^5 + (2)\,(3)\,(7)^7\,t^3$$
$$+ (2)\,(7)^5\,(31)\,t^2 + (2)\,(7)^3\,(83)\,t - (2)^2\,(7)^2\,(11)\,(17)$$

```
> # GROUP 5, IDENTITY 1: U_B4(p1*t^(-1))
  m := 5 :
  S4_5_1 := series(B4q * p1q / tq, q, 7 * T) :
  target4_5_1 := add(coeff(S4_5_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R4_5_1 := expand(findlincombo(target4_5_1, basis4, symbols4, q, 0)) :
  collect(R4_5_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$((2)^3\,(7)^8\,t^4 + (2)^4\,(5)\,(7)^6\,t^3 + (2)^3\,(3)^3\,(7)^4\,t^2 + (2)^3\,(3)\,(7)\,(23)\,t)\,p0 + ((7)^{11}\,t^5 \qquad (29)$$
$$+ (7)^9\,(19)\,t^4 + (2)\,(3)^2\,(7)^8\,t^3 + (3)\,(7)^5\,(109)\,t^2 + (2)\,(7)^4\,(17)\,t + (29))\,p1 - 1$$
$$+ (7)^{10}\,t^5 + (7)^8\,(11)\,t^4 + (2)\,(7)^6\,(19)\,t^3 + (7)^4\,(31)\,t^2 + (7)\,(41)\,t$$

```
> # GROUP 5, IDENTITY 2: U_B4(p1*t^(-2))
  m := 5 :
  S4_5_2 := series(B4q * p1q / tq^2, q, 7 * T) :
  target4_5_2 := add(coeff(S4_5_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
```

```
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_5_2 := expand(findlincombo(target4_5_2, basis4, symbols4, q, 0)) :
collect(R4_5_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^3\,t^2-(2)^3\,(7)^2\,t\right)p0+\left((2)^3\,(7)^2\,t+(7)^2\right)p1-(7)^3\,t^2-(2)^3\,(7)^2\,t+(11) \quad (30)$$

```
> # GROUP 5, IDENTITY 3: U_B4(p1*t^(-3))
m := 5 :
S4_5_3 := series(B4q * p1q / tq^3, q, 7 * T) :
target4_5_3 := add(coeff(S4_5_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_5_3 := expand(findlincombo(target4_5_3, basis4, symbols4, q, 0)) :
collect(R4_5_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^5\,t^3-(2)^3\,(7)^4\,t^2\right)p0+\left((2)^3\,(7)^4\,t^2+(7)^4\,t\right)p1-(7)^5\,t^3-(2)^3\,(7)^4\,t^2 \quad (31)$$
$$-(7)^4\,t-(2)^5\,(3)$$

```
> # GROUP 5, IDENTITY 4: U_B4(p1*t^(-4))
m := 5 :
S4_5_4 := series(B4q * p1q / tq^4, q, 7 * T) :
target4_5_4 := add(coeff(S4_5_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_5_4 := expand(findlincombo(target4_5_4, basis4, symbols4, q, 0)) :
collect(R4_5_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^4\,(3)\,(7)^7\,t^4+(2)^3\,(3)^2\,(7)^6\,t^3+(2)^3\,(3)^3\,(7)^4\,t^2+(2)^4\,(5)\,(7)^2\,t+(2)^3\right)p0+\left(\right. \quad (32)$$
$$\left.-(2)^4\,(3)\,(7)^6\,t^3-(5)\,(7)^5\,(13)\,t^2-(2)^3\,(3)\,(7)^4\,t-(2)^5\,(7)\right)p1-(7)^7\,t^4$$
$$+(2)^4\,(7)^6\,t^3+(7)^4\,(199)\,t^2+(2)^3\,(7)^2\,(83)\,t+(2)^2\,(5)\,(37)$$

```
> # GROUP 5, IDENTITY 5: U_B4(p1*t^(-5))
m := 5 :
S4_5_5 := series(B4q * p1q / tq^5, q, 7 * T) :
target4_5_5 := add(coeff(S4_5_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
```

```
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_5_5 := expand(findlincombo(target4_5_5, basis4, symbols4, q, 0)) :
collect(R4_5_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\begin{aligned}&(-(2)^3\,(7)^9\,t^5-(2)^6\,(7)^8\,t^4-(2)^4\,(5)\,(7)^7\,t^3-(2)^3\,(3)^3\,(7)^5\,t^2-(2)^4\,(5)\,(7)^3\,t\\&-(2)^3\,(7))\,p0+((2)^3\,(7)^8\,t^4+(3)^2\,(7)^8\,t^3+(2)^3\,(3)^2\,(7)^6\,t^2+(2)^3\,(3)\,(7)^5\,t\\&+(2)^5\,(7)^2)\,p1-(7)^9\,t^5-(2)^3\,(7)^8\,t^4-(7)^7\,(31)\,t^3-(2)^3\,(7)^5\,(31)\,t^2\\&-(2)^3\,(7)^3\,(83)\,t-(2)^2\,(5)\,(7)\,(37)\end{aligned}\qquad (33)$$

```
> # GROUP 5, IDENTITY 6: U_B4(p1*t^(-6))
m := 6 :
S4_5_6 := series(B4q * p1q / tq^6, q, 7 * T) :
target4_5_6 := add(coeff(S4_5_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_5_6 := expand(findlincombo(target4_5_6, basis4, symbols4, q, 0)) :
collect(R4_5_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&(-(2)^3\,(7)^{11}\,t^6-(2)^3\,(7)^{10}\,t^5+(2)^3\,(5)\,(7)^9\,t^4+(2)^4\,(5)^2\,(7)^7\,t^3+(2)^3\,(3)^3\,(5)\,(7)^5\,t^2\\&+(2)^4\,(5)^2\,(7)^3\,t+(2)^3\,(5)\,(7))\,p0+((2)^3\,(7)^{10}\,t^5+(7)^{10}\,t^4-(2)^3\,(5)\,(7)^8\,t^3\\&-(2)^3\,(3)^2\,(5)\,(7)^6\,t^2-(2)^3\,(3)\,(5)\,(7)^5\,t-(2)^5\,(5)\,(7)^2)\,p1-(7)^{11}\,t^6\\&-(2)^3\,(7)^{10}\,t^5-(7)^{10}\,t^4+(2)^3\,(3)\,(5)\,(7)^7\,t^3+(2)^3\,(5)\,(7)^5\,(31)\,t^2\\&+(2)^3\,(5)\,(7)^3\,(83)\,t+(2)^4\,(5)\,(7)\,(59)\end{aligned}\qquad (34)$$

```
> # GROUP 5, IDENTITY 7: U_B4(p1*t^(-7))
m := 8 :
S4_5_7 := series(B4q * p1q / tq^7, q, 7 * T) :
target4_5_7 := series(tq * (add(coeff(S4_5_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T)), q, T) :
basis4 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R4_5_7 := expand(findlincombo(target4_5_7, basis4, symbols4, q, 0) / t) :
collect(R4_5_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$

$$\#\ of\ terms\ , 48$$

$$(-(2)^3\,(7)^{13}\,t^7-(2)^3\,(7)^{12}\,t^6-(2)^3\,(7)^9\,(17)\,t^4-(2)^4\,(5)\,(7)^7\,(17)\,t^3 \qquad (35)$$

$$-(2)^3\,(3)^3\,(7)^5\,(17)\,t^2-(2)^4\,(5)\,(7)^3\,(17)\,t-(2)^3\,(7)\,(17))\,p0+\Big((2)^3\,(7)^{12}\,t^6$$

$$+(2)\,(3)\,(7)^{11}\,t^5-(7)^9\,(19)\,t^4+(2)\,(7)^7\,(409)\,t^3+(3)\,(7)^6\,(389)\,t^2$$

$$+(2)\,(7)^4\,(1399)\,t+(7)^4\,(11)+\frac{1}{t}\Big)\,p1+(2)\,(3)\,(7)^{13}\,t^7+(2)^2\,(5)\,(7)^{12}\,t^6$$

$$+(2)^4\,(7)^{10}\,(17)\,t^5+(3)\,(7)^8\,(631)\,t^4+(3)\,(5)\,(7)^7\,(29)\,t^3-(7)^5\,(2953)\,t^2$$

$$-(2)^2\,(3)^2\,(5)\,(7)^3\,(59)\,t-(7)\,(151)\,(179)$$

```
> # GROUP 6, IDENTITY 1: U_B4(1)
  m := 1 :
  S4_6_1 := series(B4q, q, 7 * T) :
  target4_6_1 := add(coeff(S4_6_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m)] :
  R4_6_1 := expand(findlincombo(target4_6_1, basis4, symbols4, q, 0)) :
  collect(R4_6_1, [t], recursive, ifactor);
```

$$nx = , 2$$

$$\#\ of\ terms\ , 23$$

$$1 \qquad (36)$$

```
> # GROUP 6, IDENTITY 2: U_B4(t^(-1))
  m := 1 :
  S4_6_2 := series(B4q / tq, q, 7 * T) :
  target4_6_2 := add(coeff(S4_6_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m)] :
  R4_6_2 := expand(findlincombo(target4_6_2, basis4, symbols4, q, 0)) :
  collect(R4_6_2, [t], recursive, ifactor);
```

$$nx = , 2$$

$$\#\ of\ terms\ , 23$$

$$-(7)\,t-(2)^2 \qquad (37)$$

```
> # GROUP 6, IDENTITY 3: U_B4(t^(-2))
  m := 2 :
  S4_6_3 := series(B4q / tq^2, q, 7 * T) :
  target4_6_3 := add(coeff(S4_6_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m)] :
  R4_6_3 := expand(findlincombo(target4_6_3, basis4, symbols4, q, 0)) :
  collect(R4_6_3, [t], recursive, ifactor);
```

$$nx = , 3$$

$$\#\ of\ terms\ ,\ 24$$

$$-(7)^3\ t^2 + (2)^2\ (5) \qquad (38)$$

```
> # GROUP 6, IDENTITY 4: U_B4(t^(-3))
  m := 3 :
  S4_6_4 := series(B4q / tq^3, q, 7 * T) :
  target4_6_4 := add(coeff(S4_6_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m)] :
  R4_6_4 := expand(findlincombo(target4_6_4, basis4, symbols4, q, 0)) :
  collect(R4_6_4, [t], recursive, ifactor);
```

$$nx = ,\ 4$$

$$\#\ of\ terms\ ,\ 25$$

$$-(7)^5\ t^3 - (2)^3\ (11) \qquad (39)$$

```
> # GROUP 6, IDENTITY 5: U_B4(t^(-4))
  m := 4 :
  S4_6_5 := series(B4q / tq^4, q, 7 * T) :
  target4_6_5 := add(coeff(S4_6_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m)] :
  R4_6_5 := expand(findlincombo(target4_6_5, basis4, symbols4, q, 0)) :
  collect(R4_6_5, [t], recursive, ifactor);
```

$$nx = ,\ 5$$

$$\#\ of\ terms\ ,\ 26$$

$$-(7)^7\ t^4 + (2)^2\ (5)\ (13) \qquad (40)$$

```
> # GROUP 6, IDENTITY 6: U_B4(t^(-5))
  m := 5 :
  S4_6_6 := series(B4q / tq^5, q, 7 * T) :
  target4_6_6 := add(coeff(S4_6_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols4 := [seq(t^j, j = 0 ..m)] :
  R4_6_6 := expand(findlincombo(target4_6_6, basis4, symbols4, q, 0)) :
  collect(R4_6_6, [t], recursive, ifactor);
```

$$nx = ,\ 6$$

$$\#\ of\ terms\ ,\ 27$$

$$-(7)^9\ t^5 + (2)^2\ (7)\ (17) \qquad (41)$$

```
> # GROUP 6, IDENTITY 7: U_B4(t^(-6))
  m := 6 :
  S4_6_7 := series(B4q / tq^6, q, 7 * T) :
  target4_6_7 := add(coeff(S4_6_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
```

```
basis4 := [seq(series(tq^j, q, T), j = 0 ..m)] :
symbols4 := [seq(t^j, j = 0 ..m)] :
R4_6_7 := expand(findlincombo(target4_6_7, basis4, symbols4, q, 0)) :
collect(R4_6_7, [t], recursive, ifactor);
```

$$nx = , 7$$

$$\#\ of\ terms\ , 28$$

$$-(7)^{11}\, t^6 - (2)^3\, (7)\, (13)\, (23) \qquad (42)$$

>